

The Racing Drift 3: Vehicle Physics

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Abstract

In this project I chose to create a racing game with the focus on coding a car that has realistic vehicle dynamics. To get the best vehicle dynamics I searched for knowledge on how the different parts of the car work and how they affect the overall handling of the car. I have also challenged myself to go beyond limits and put my own custom code into the scripts.

The part I put most time and focus on was the wheels. Although the wheels of a car may appear simple, they are really one of the most complex parts of a car as all of the car's forces goes through the wheels

The result of this is a car that responds in relatively believable ways with consistent predictable handling especially during drifting, which was a major focus of the gameplay experience.

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1. Introduction

A lot of gamers like to play racing games and for me it is a challenge to develop a racing game as realistic as possible compared to a real racing car.

The project has focused on wheel physics in the car game The Racing Drift 3.

The project shows car physics with forces, friction and drifting on a track for demonstration. Everything that makes the car move and handle is dependent on engine, wheels, suspension, anti-sway bar in a real car. As all the forces are combined and co-operates to get the best result.

In my results, chapter 3, I will describe the technical implementation of my scripts.

2. Methodology

The project started with thorough investigation about Individual car parts and how they work.

To get realistic physics in the project i studied two YouTube videos, one wheel tutorial “Making Custom Car Physics in Unity (for Very Very Valet)”¹ and the one describing how wheels work in reality “How do Race Car Tyres/Tires Work? Slip Angle, Contact Patch Deformation etc.”².

Three articles “Introduction to Vehicle dynamics and suspension systems”³ for overall physics, “Anti-roll bar”⁴ for more in depth on the anti-roll bar and “Automobile handling”⁵ for tuning help.

For the game, Unity was chosen as the game engine to save time as I have more lecture material in unity, C# and experience with Unity. I used the unity's animation curve for engine and wheels. Suspension and anti-sway bar use only float variables.

In my planning I decided to start ahead of schedule to have extra time for my report. Meanwhile I wrote my report I polish the code and values in the unity inspector for the car.

¹ <https://www.youtube.com/watch?v=CdPYlj5uZel>

² <https://www.youtube.com/watch?v=y5Y-w4zGW00>

³ <https://www.school-mechademic.com/blog/vehicle-dynamics>

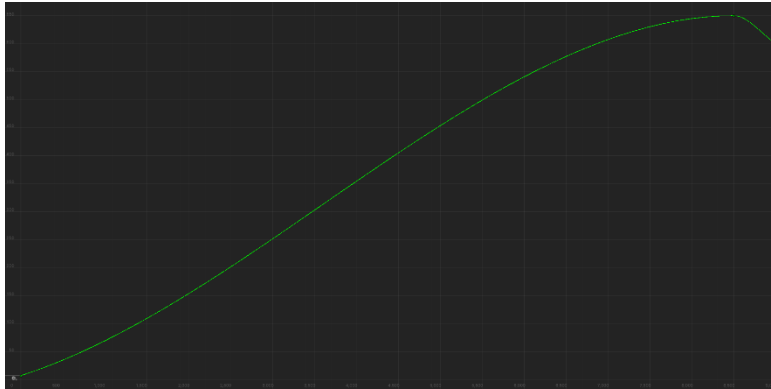
⁴ https://en.wikipedia.org/wiki/Anti-roll_bar

⁵ https://en.wikipedia.org/wiki/Automobile_handling#Aftermarket_modifications_and_adjustments

3. Result

3.1 Engine

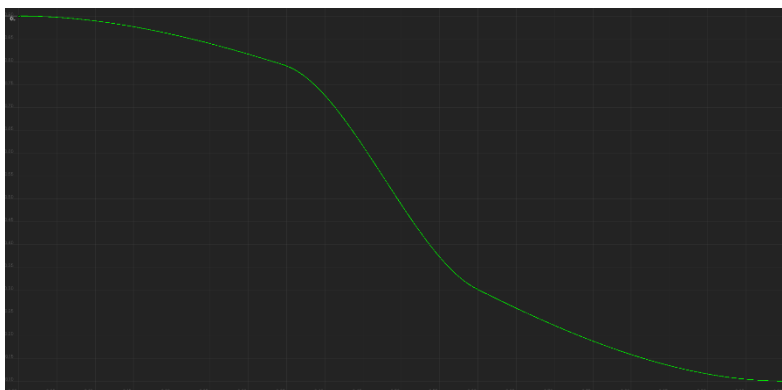
The engine is an animation curve where the x axis is the rpm, and the y axis is the HP which gives possibility of different kinds of engines. By making the curve flat you get an electric engine. The engine in the car is gas powered to have a more dynamic engine curve. That gives a ramp up to the max HP.



3.2 Wheel

Real tires have a frictional ellipse⁶ that describes their friction in longitude and latitudinal acceleration. I had to use two friction curves, one for longitude and the other for latitudinal acceleration.

The friction curve used is not realistic as the curve goes from high to low while true friction curves⁷ for tires climb in friction coefficient to a peak and then goes back down and flattens out. The wheel slip is not perfect as all gears will make the wheels slip and spin when accelerating.



⁶<https://www.youtube.com/watch?v=y5Y-w4zGW00>

⁷<https://www.youtube.com/watch?v=y5Y-w4zGW00>

When the car no longer has the force necessary to sustain the drift and the car doesn't point in the exact direction of travel the cars rotation snaps out of the drift, so the front is in the direction of travel.

By linearly interpolating the friction from the current friction multiplier to the standard friction multiplier it gets smoothed out.

3.3 Suspension

The suspension uses the wheel origin as the connection to the car and then checks the rest distance to the ground. All you need for the force of the spring is the offset from the rest distance times the strength of the spring and you got a spring that wants to move back to the rest distance. To stop it you need a dampening force which is the negative of the velocity times dampening.

For offset we grab the rest distance minus the distance to the ground and for the velocity we grab the cars upward velocity at the wheels position.

3.4 Anti-Sway Bar

An anti-sway bar is a torsion spring that connects the left and right suspension. When the left suspension gets compressed, it applies force to the torsion spring which in turn compresses the right suspension.

3.5 Tuning & Value Tweaking

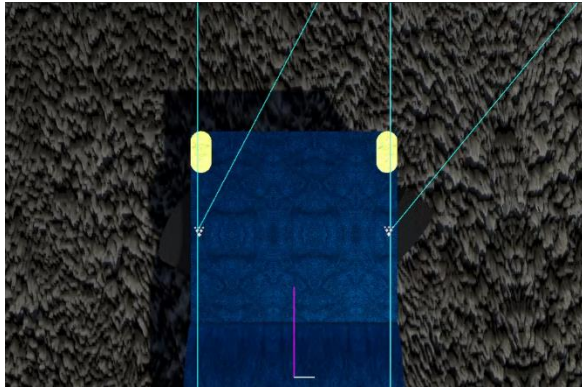
The car has the possibility to get any type of drive train (rwd, fwd, awd) but that is not an implemented setting. In the project I tuned the car for rear wheel drive and worked on finding the best tuning.

The further forward the center of mass is, the more stable the car gets (not wanting to drift) and the further back the center of mass is the more it wants to drift and slide, When turning at high speeds the mass has the tendencies to kick up the front most notably with fwd and awd drive.

Camber helps with steering but with how the wheel resists sideways sliding puts torque on the car. Negative camber pushes up on the inside of the turn and positive pushes down on the inside. Here the anti-sway bar would be perfect as it would lift the opposite wheel and stop the car from rolling over.

The toe helps tunes steering toe in decrease steering and toe out increase turning. But toe will decrease the speed of the car as the wheels won't be pointing in the direction of travel.

Perfect Ackermannⁱ meaning that both left and right wheels with steering turn differently not taking in any external factors such as slip angle, **outside wheel turn less than the inside wheel** to match the curves they follow in the corner.



3.6 Track

The track is put together on a giant plain 11 high speed, 3 medium speed and one hairpin corners. The width of the road is consistent throughout the entire track. It is made to test the car's capability. There are in total 11 checkpoints, but the first checkpoint is not counted in to the UI as it is considered as the goal, but the function is the same. Lap count can be set through the options in the main menu.

There is no extra resistance from driving of the track as the track is more of a reference point to see the behavior of the car.

4. Discussion

4.1 Engine

I made an animation curve for a combustion engine. The curve shows the horsepower which affects acceleration and top speed.

By making the engine an animation curve it becomes easy to add other engine profiles.

The curve helps with setting the right speed in a realistic way for the car.

I took code from a previous project and added code from a previous lecture. The result was that I got more realistic shifting compared to my previous project and a more dynamic engine compared to the lecture.

I made a turbo but did not have a noticeable effect which could be because of wrong calculations or size and power of it. I went on with plans to remake the engine and use

virtual cylinders with volume. That would help me calculate the effect of the turbo using the volume of the cylinders. With the time limit it might have been too much, so I scraped it.

4.2 Wheel

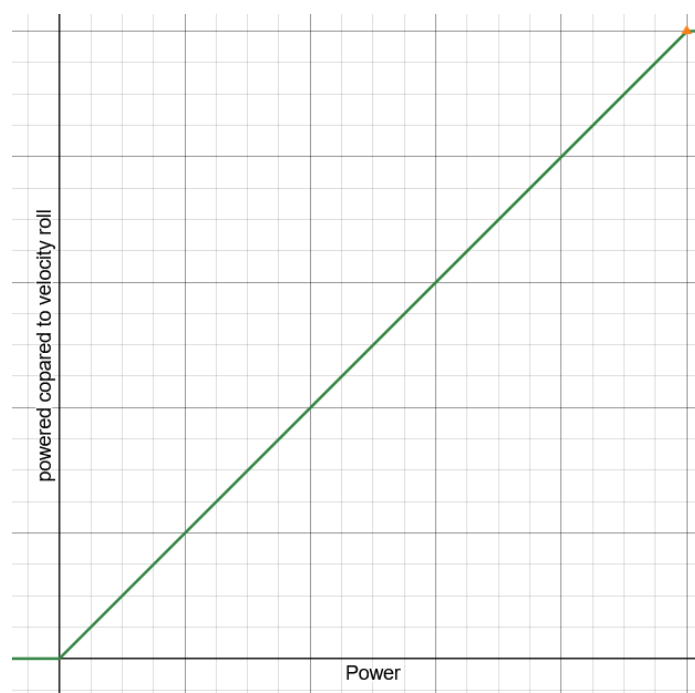
I began with a code from Making Custom Car Physics in Unity (for Very Very Valet)ⁱⁱ. Then I added two animation curves for friction. It makes the friction rely on both forward and sideways slip. I added a multiplier to increase the grip as default and lower it when drifting. These two animation curves together and the multiplier make the drifting more realistic.

I scooped down and compromised on the wheel's friction curve and wheelspin because the return I got from that became too small.

To visually simulate the rotation of the wheels, I calculate the wheel's rotation based on both the wheel's current velocity and the power from the engine. First, I handle each component separately. Then, I combine them by subtracting the wheel's velocity from the engine's rotational speed, clamping the result between 0 and the engine's maximum rotation, and finally, I add the wheel's velocity back into the equation. This is represented by the formula $(f_{curve1}(x) = (V + f_{clamp}(x - V, 0, x)))$.

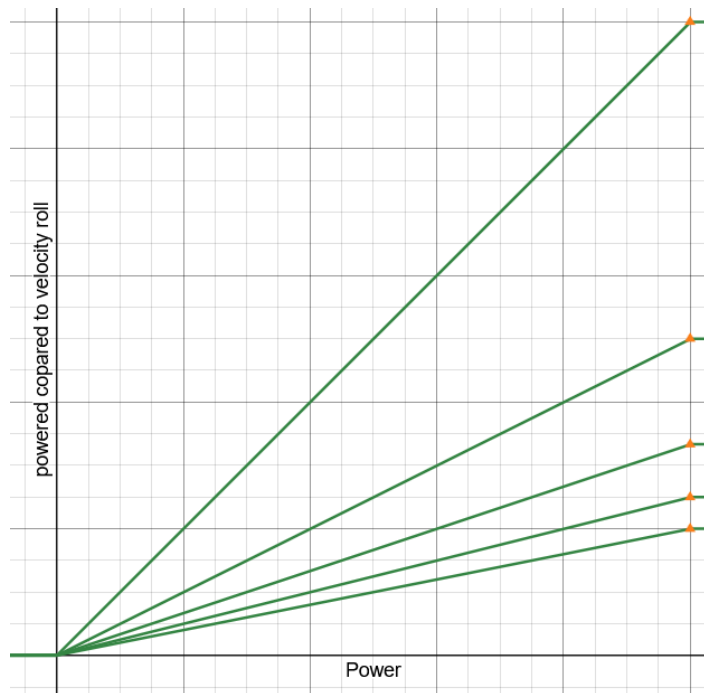
Where V is the velocity, x is the engines rotational speed, f_{clamp} ensures that the difference between the engine roll and the wheel's velocity is bounded between 0 and engines rotational speed.

This approach gives me a dynamic wheel system, where the wheel's rotation is influenced by the engine's power when the throttle is applied, and by the wheel's own velocity when the throttle is not engaged.



Clamped the value to be max 10 for the graph

But this gives us a small issue. The powered wheels will always be slipping with the same amount no matter the velocity or the torque from the gear. To solve this, I divided the formula by the current gear before adding back the velocity. Which gave the following formula $f_{curve2}(x) = (V + (f_{clamp}(x - V, 0, x) / G))$. Where G is the current gear. This will make the slipping of the powered wheels less in higher gears



Top line is first gear and bottom line is fifth gear

Clamped the value to be max 10 for the graph

4.3 Suspension

The weight of the car gets balanced on the springs while turning. The handling of the car with the suspension is more realistic as the mass of the car will shift when turning to the outside wheels causing the car to tilt.

I got the suspension code from Making Custom Car Physics in Unity (for Very Very Valet) but I made a key modification. Instead of using the wheel's up vector to determine the direction of the spring, I changed it to align with the car's up vector. This adjustment was necessary to compensate for the wheel camber. If not, the camber would angle the suspension and kick up the front of the car in an unnatural way while turning.

4.4 Anti-Sway Bar

Taking inspiration from another game that allows the player to change the stiffness of the anti-sway bar of the car, I decided to try to make my own.

The attempt to improve the car's handling by creating code to mimic an anti-sway bar was not successful. Instead of bringing up the opposite side spring it just adds force on the opposite side when it should bring it up a percentage value. So, I turned the strength of it down to zero making it have no effect.

4.5 Tuning & Value Tweaking

By making important values, (for example FrictionMultiplier, SteeringRadius, etc) that could need to be changed serializable by Unity, they become visible and editable in Unity. This makes it easy to change the handling of the car. This also made it possible to change the values on the fly letting me find the best values for realism.

I did not test or code anti Ackerman⁸ which is opposite of perfect Ackerman. Anti Ackerman turns the outside wheels more than the inside wheels. It is mostly used in F1 as it works best in the high-speed corners as it is worse for tight cornering. The game can simulate it in corners by increasing toe and this decreases the turn radius as the inner wheel of the corner is best at applying force to rotate the car and that is why anti Ackerman is barely used.

4.6 Track

I made the track with three types of corners, high speed, medium speed and hairpin corners. It also tests the handling of the car, turning, drifting, braking, speed etc.

I added checkpoints strategically at reasonable distances.

To get out more of the track i should have made the track less wide and added another hairpin.

5. Conclusion

To make the Project realistic I have to have knowledge of how cars work.

It has been a challenge to get the wheels right for the game as the Wheels are one of the most complex parts of a car.

The base code of the wheels can be used for several objects with directional movement and resisting movement in other directions.

If I continue, I have the possibility to develop the anti-sway bar, add toe, add multiple cars with different settings.

⁸ <https://www.youtube.com/watch?v=O2ONv5MrpFg>

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ⁱ <https://www.youtube.com/watch?v=O2ONv5MrpFg>

ⁱⁱ <https://www.youtube.com/watch?v=CdPYlj5uZel>